

AI Inventory Forecasting Platform

Overview

As part of my Bachelor of Computer and Information Sciences, I led a five-person development team in designing and implementing an AI-powered inventory forecasting platform for a national retail client. The objective was to replace a manual stock ordering process with an intelligent forecasting system capable of predicting future inventory demand across multiple retail locations.

The Problem

The client relied on managers manually estimating stock requirements every two weeks. This process was time-consuming, inconsistent, and frequently resulted in overstocking or stock shortages despite having years of detailed historical sales data available. The project transformed this historical data into actionable inventory forecasts using machine learning.

My Role

Team Manager & Data Analyst

- Managed a five-person development team
- Performed exploratory data analysis and preprocessing
- Engineered forecasting features
- Trained and evaluated multiple forecasting models
- Selected and optimised the final LightGBM model
- Designed the forecasting dashboard
- Presented findings and recommendations to stakeholders

Technical Solution

Developed an end-to-end machine learning pipeline including data extraction, preprocessing, feature engineering, temporal train/test splitting, model evaluation, hyperparameter optimisation, LightGBM multi-horizon forecasting, and an interactive dashboard displaying 14-day inventory forecasts.

Technologies Used

Python, LightGBM, Scikit-learn, Pandas, NumPy, Matplotlib, Flask, PostgreSQL, Visual Studio Code, Jupyter Notebook and Git.

Business Impact

Delivered automated inventory forecasting, improved stock allocation recommendations, reduced overstocking and understocking risk, scalable forecasting across multiple retail locations, live database integration, and an intuitive operational dashboard.

Final Model Performance

Metric	Result
MAE	0.0935
RMSE	0.4380
MAPE	6.40%

Key Skills Demonstrated

Machine Learning, Time Series Forecasting, Feature Engineering, Data Cleaning & Preprocessing, Model Evaluation, Hyperparameter Optimisation, Python Development, Database Integration, Dashboard Development, Agile Team Leadership, Client Communication and Technical Documentation.